

A **MERN Stack Developer** is a skilled programmer who specializes in building web applications using four key technologies: **MongoDB, Express, React, and Node.js**. These technologies work together to create both the front-end and back-end of a website.

What's React.JS:

- ReactJS is a popular JavaScript library for creating user interfaces of web applications. It helps developers build reusable components. ReactJS is known for its efficiency and renders components quickly.

What's MongoDB:

- MongoDB is a **NoSQL database** that stores data in a flexible, **JSON-like format** instead of tables. It is designed for **scalability, high performance, and ease of development**.

MongoDB VS SQL DB

Feature	SQL (MySQL, PostgreSQL, etc.)	MongoDB (NoSQL)
Data Storage	Stores data in tables (rows & columns)	Stores data in documents (JSON format)
Structure	Fixed Schema – Every row must have the same fields	Flexible Schema – Each document can have different fields
Scalability	Scales vertically (more CPU/RAM)	Scales horizontally (more servers)
Performance	Good for complex queries & transactions	Fast for big data & real-time apps
Use Cases	Banking, ERP, CRM (structured data)	IoT, Big Data, social media, AI apps (unstructured data)

In simple terms:

- **SQL** = Organized, structured, traditional, great for transactions.
- **MongoDB** = Flexible, scalable, great for fast-growing, dynamic applications.

MongoDB CRUD Operations (Create, Read, Update, Delete)

Create = insertOne(), insertMany()

Read = findOne(), find()

Update = updateOne(), updateMany()

Delete = deleteOne(), deleteMany()

SQL CRUD Operations (Create, Read, Update, Delete)

Create = INSERT INTO

Read = SELECT

Update = UPDATE

Delete = DELETE

What's MongoDB:

- **Express.js** is a backend web framework for Node.js that makes it easier to build web servers and APIs. Think of Node.js as a car engine. Express.js is like a dashboard that makes it easier to drive

What's NodeJS:

- Node.js is a JavaScript runtime that lets you run JavaScript outside the browser. It is used to build fast and scalable backend applications. Think of JavaScript as a chef, and Node.js as the kitchen where the chef can cook not just in the browser but also on the server!

STAR Method

The **STAR method** is a structured way to answer **behavioural interview questions**. It helps you explain **real-life experiences** clearly and effectively.

- **S → Situation** (What was the challenge or task?)
T → Task (What was your role or responsibility?)
A → Action (What steps did you take?)
R → Result (What was the outcome?)

E.g.

1. Describe a situation where you worked under pressure as a developer.

S – Situation:

"As a junior developer at my first job, I was assigned to fix a critical bug in a production system just a week after joining. The issue was causing errors for live users, and the senior developer was unavailable."

T – Task:

"I had to quickly understand the existing codebase, identify the root cause, and implement a fix under tight deadlines while ensuring the system remained stable."

A – Action:

"I first analyzed the error logs and traced the issue to a faulty API call. I reached out to my team for guidance, researched similar problems, and used debugging tools to isolate the bug. I implemented a fix, tested it thoroughly, and deployed it carefully to minimize downtime."

R – Result:

"The fix was successful, and system errors dropped immediately. My manager appreciated my quick problem-solving, and the experience helped me gain confidence in debugging and working under pressure."

If you're failing to meet a deadline, how do you communicate:

If I'm falling behind on a deadline, I first analyse the issue to understand why. Then, I communicate with my manager early, prioritize the most important tasks, and adjust my plan to focus on delivering the critical parts first. If needed, I ask for help or suggest a realistic new timeline. Afterward, I reflect on what went wrong to improve my time management for future tasks.